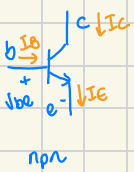
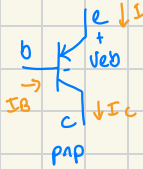
ELE 2136

Lecture 6

Intro to BJT & DC analysis



npn



pnp

We want BJT to work as amplifiers

→ To achieve that they need to be in the active mode

Active mode conditions:

npn: $V_{BE} = V_{D0} \approx 0.7\,[V]$, $V_C > V_B$

pnp: $V_{EB} = V_{D0} \approx 0.7\,[V]$, $V_B > V_C$

$$I_C = \beta I_B$$

↑ current gain

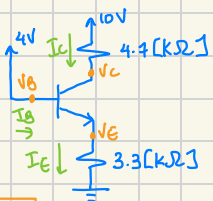
$$I_E = \left(\frac{\beta+1}{\beta}\right) I_C$$

$$I_E = \left(\frac{\beta+1}{\beta}\right)\beta I_B \Rightarrow \boxed{I_C = (\beta+1) I_B}$$

DC analysis

→ find $V_C, V_B, V_E, I_C, I_B, I_E$

Example: Given $V_{D0} = 0.7\,[V]$, $\beta = 100$



$$I_E = \frac{V_E - 0}{3300} = \frac{3.3}{3300} = 0.001\,[A] = 1\,[mA]$$

$$\Rightarrow \boxed{I_E = 0.001\,[A] = 0.1\,[mA]}$$

$$I_E = \left(\frac{\beta+1}{\beta}\right) I_C \Rightarrow I_C = \left(\frac{\beta}{\beta+1}\right) I_E \Rightarrow I_C = \left(\frac{100}{100+1}\right)(0.001) = 0.000990\,[A]$$

$$\Rightarrow \boxed{I_C = 0.000990\,[A] = 0.99\,[mA]}$$

$V_B = 4\,[V]$

$V_{BE} = V_B - V_E = V_{D0} = 0.7$

$\Rightarrow 4 - V_E = 0.7$

$\Rightarrow 4 - 0.7 = V_E \Rightarrow \boxed{V_E = 3.3\,[V]}$

$$I_C = \beta I_B \Rightarrow I_B = \frac{I_C}{\beta} = \frac{0.000990}{100} = 9.9\times10^{-6}\,[A] = 9.9\,[\mu A]$$

$$\Rightarrow \boxed{I_C = 9.9\times10^{-6}\,[A] = 9.9\,[\mu A]}$$

$$I_C = \frac{10 - V_C}{4700}$$

$$\Rightarrow 4700 I_C = 10 - V_C \Rightarrow V_C = 10 - 4700 I_C \Rightarrow V_C = 10 - 4700(9.9\times10^{-6})$$

$$\Rightarrow \boxed{V_C = 9.95\,[V]}$$

More complex example with resistive network

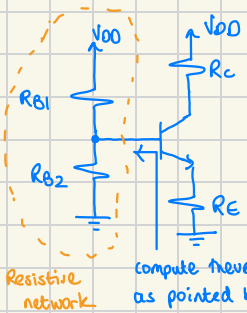
$\beta = 100$, $V_{DD} = 12V$

$R_{B1} = 30[k\Omega]$

$R_{B2} = 10[k\Omega]$

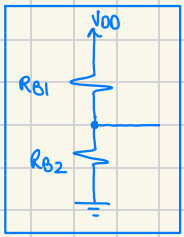
$R_C = 4.3[k\Omega]$

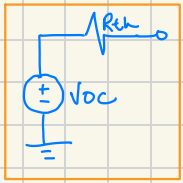
$R_E = 1.3[k\Omega]$

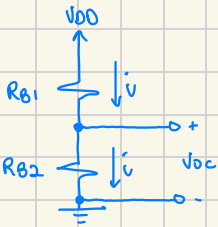


compute thevenin equivalent circuit as pointed by the arrow.

Replacing this with this

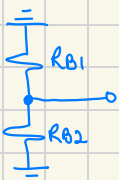






Voltage divider: $V_{OC} = \left(\frac{R_{B2}}{R_{B1}+R_{B2}}\right) V_{DD}$

Finding Rth: disable all independent sources

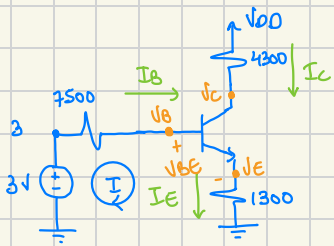


$\Rightarrow R_{th} = R_{B1} // R_{B2}$.

$\Rightarrow V_{OC} = 3[V]$

$R_{th} = 7500[\Omega]$

Overall circuit becomes as such:



KVL @ I: $-3 + 7500 I_B + V_{BE} + 1300 I_E = 0$, $V_{BE} = 0.9[V]$

$\Rightarrow -3 + 7500 I_B + 0.7 + 1300 I_E = 0$

$\Rightarrow -2.3 + 7500 I_B + 1300 I_E = 0$

$I_B = \frac{1}{\beta} I_C$, $I_C = \left(\frac{\beta}{\beta+1}\right) I_E$

$\Rightarrow I_B = \frac{1}{\beta}\left(\frac{\beta}{\beta+1}\right) I_E \Rightarrow \boxed{I_B = \frac{1}{101} I_E}$

$\Rightarrow -2.3 + \frac{7500}{101} I_E + 1300 I_E = 0$

$\Rightarrow \left(\frac{7500}{101} + 1300\right) I_E = 2.3 \Rightarrow I_E = \frac{2.3}{\left[\frac{7500}{101} + 1300\right]} = 0.00167[A]$

$I_C = \left(\frac{\beta+1}{\beta}\right) I_E = \left(\frac{101}{100}\right)(0.00164)$

$\Rightarrow \boxed{I_C = 0.00169[A] = 1.69[mA]}$

$\Rightarrow I_B = \frac{I_C}{\beta} = \frac{0.00169}{100} \Rightarrow \boxed{I_B = 0.00001690 4[A]}$

$I_B = \frac{3 - V_B}{7500} \Rightarrow 7500(0.00001690 4) = 3 - V_B$

$\Rightarrow \boxed{V_B = 2.87[V]}$

$\Rightarrow \boxed{I_E = 0.00167[A] = 1.67[mA]}$

$V_{BE} = V_B - V_E = 0.7 \Rightarrow 2.87 - V_E = 0.7 \Rightarrow \boxed{V_E = 2.17[V]}$

$I_C = \frac{12 - V_C}{4300} \Rightarrow 4300(0.00169) = 12 - V_C \Rightarrow \boxed{V_C = 4.73[V]}$

Check active mode condition

$V_{BE} = 2.87 - 2.17 = 0.7\,[V]$ ✓

$V_C > V_B \Rightarrow 4.73 > 2.87$ ✓

} ⇒ Active mode accepted.